

Chapter 8 Momentum and forces

Section 8.1 Momentum and conservation of momentum

Worked example: Try yourself 8.1.1

MOMENTUM

Calculate the momentum of a 1230 kg car driving at 16.7 ms^{-1} north.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 1230 \text{ kg}$ $v = 16.7 \text{ ms}^{-1}$ north
Apply the equation for momentum.	$p = mv$ $= 1230 \times 16.7$ $= 20541$ $= 20500 \text{ kg ms}^{-1}$
Ensure that the final answer is in the same direction as the velocity.	$p = 20500 \text{ kg ms}^{-1}$ north

Worked example: Try yourself 8.1.2

CONSERVATION OF MOMENTUM

A 1200 kg wrecking ball is moving at 2.50 ms^{-1} north towards a 1500 kg wrecking ball moving at 4.00 ms^{-1} south. Calculate the final velocity of the 1500 kg ball if the 1200 kg ball rebounds at 3.50 ms^{-1} south.	
Thinking	Working
Identify the variables using subscripts. Ensure that the variables are in their standard units.	$m_1 = 1200 \text{ kg}$ $u_1 = 2.50 \text{ ms}^{-1}$ north $v_1 = 3.50 \text{ ms}^{-1}$ south $m_2 = 1500 \text{ kg}$ $u_2 = 4.00 \text{ ms}^{-1}$ south $v_2 = ?$
Apply the sign convention to the variables.	$m_1 = 1200 \text{ kg}$ $u_1 = +2.50 \text{ ms}^{-1}$ $v_1 = -3.50 \text{ ms}^{-1}$ $m_2 = 1500 \text{ kg}$ $u_2 = -4.00 \text{ ms}^{-1}$ $v_2 = ?$
Apply the equation for conservation of momentum.	$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$ $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(1200 \times 2.50) + (1500 \times -4.00) = (1200 \times -3.50) + 1500 v_2$ $1500 v_2 = 3000 + -6000 - (-4200)$ $v_2 = \frac{1200}{1500}$ $= 0.800 \text{ ms}^{-1}$
Apply the sign convention to describe the direction of the final velocity.	$v_2 = 0.80 \text{ ms}^{-1}$ north

Worked example: Try yourself 8.1.3
CONSERVATION OF MOMENTUM WHEN MASSES COMBINE

An 80.0 kg rugby player is moving at 1.50 m s^{-1} north when he tackles an opponent with a mass of 50.0 kg who is moving at 5.00 m s^{-1} south. Calculate the final velocity of the two players.	
Thinking	Working
Identify the variables using subscripts and ensure that the variables are in their standard units. Add m_1 and m_2 to get m_3 .	$m_1 = 80.0 \text{ kg}$ $u_1 = 1.50 \text{ m s}^{-1}$ north $m_2 = 50.0 \text{ kg}$ $u_2 = 5.00 \text{ m s}^{-1}$ south $m_3 = 130 \text{ kg}$ $v_3 = ?$
Apply the sign convention to the variables.	$m_1 = 80.0 \text{ kg}$ $u_1 = +1.50 \text{ m s}^{-1}$ $m_2 = 50.0 \text{ kg}$ $u_2 = -5.00 \text{ m s}^{-1}$ $m_3 = 130 \text{ kg}$ $v_3 = ?$
Apply the equation for conservation of momentum.	$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$ $m_1 u_1 + m_2 u_2 = m_3 v_3$ $(80 \times 1.50) + (50 \times -5.00) = 130 v_3$ $v_3 = \frac{120 + -250}{130}$ $= -1.00 \text{ m s}^{-1}$
Apply the sign to describe the direction of the final velocity.	$v_3 = 1.00 \text{ m s}^{-1}$ south

Worked example: Try yourself 8.1.4
CONSERVATION OF MOMENTUM FOR EXPLOSIVE COLLISIONS

A 2000 kg cannon fires a 10 kg cannonball. The cannon and the cannonball are initially stationary. After firing, the cannon recoils with a velocity of 8.15 m s^{-1} north. Calculate the velocity of the cannonball just after it is fired.	
Thinking	Working
Identify the variables using subscripts and ensure that the variables are in their standard units. Note that m_1 is the sum of the bodies i.e. the cannon and the cannonball.	$m_1 = 2010 \text{ kg}$ $u_1 = 0 \text{ m s}^{-1}$ $m_2 = 2000 \text{ kg}$ $u_2 = 8.15 \text{ m s}^{-1}$ north $m_3 = 10 \text{ kg}$ $v_3 = ?$
Apply the sign convention to the variables.	$m_1 = 2010 \text{ kg}$ $u_1 = 0 \text{ m s}^{-1}$ $m_2 = 2000 \text{ kg}$ $u_2 = +8.15 \text{ m s}^{-1}$ $m_3 = 10 \text{ kg}$ $v_3 = ?$

Apply the equation for conservation of momentum for explosive collisions.	$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$ $m_1 u_1 = m_2 u_2 + m_3 v_3$ $2010 \times 0 = (2000 \times 8.15) + 10v_3$ $v_3 = \frac{0 - 16300}{10}$ $\frac{-16300}{10}$ $= -1630 \text{ ms}^{-1}$
Apply the sign to describe the direction of the final velocity.	$v_3 = 1630 \text{ ms}^{-1}$ south

Section 8.1 Review

KEY QUESTIONS SOLUTIONS

- $p = mv$
 $= 3.50 \times 2.50$
 $= 8.75 \text{ kg ms}^{-1}$ south
- $p = mv$
 $= 433 \times 22.2$
 $= 9612.6$
 $= 9610 \text{ kg ms}^{-1}$ west
- $p = mv$
 $= 0.065 \times 61.0$
 $= 3.97 \text{ kg ms}^{-1}$ south
- First ball: $p = mv = 4.5 \times 3.5 = 15.75 \text{ kg ms}^{-1}$
 Second ball: $p = mv = 2.5 \times 6.8 = 17 \text{ kg ms}^{-1}$
 The second ball has the greater momentum.
- $$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(70.0 \times 0) + (400 \times 0) = (70.0 \times 2.50) + 400v_2$$

$$400v_2 = 0 + -175$$

$$v_2 = \frac{-175}{400}$$

$$= -0.438 \text{ ms}^{-1}$$

The boat moves backwards at 0.438 ms^{-1} (to 3 significant figures).
- $$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(0.070 \times 0) + (0.545 \times 80) = (0.070 \times 75.0) + 0.545v_2$$

$$0.545v_2 = 0 + 43.6 - 5.25$$

$$v_2 = \frac{38.35}{0.545}$$

$$= 70.4 \text{ ms}^{-1}$$
 (to 3 significant figures)
- $$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_3 v_3$$

$$(2500 \times 2.00) + (m_2 \times 0) = (2500 + m_2) \times 0.300$$

$$5000 = 0.300 \times 2500 + 0.300m_2$$

$$0.300m_2 = 5000 - 750$$

$$m_2 = \frac{4250}{0.300}$$

$$= 14167$$

$$= 14200 \text{ kg}$$
 (to 3 significant figures)

- 8 $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$
 $m_1 u_1 = m_2 v_2 + m_3 v_3$
 $0 = (9995)v_2 + (5.0 \times 6000)$
 $v_2 = \text{velocity of space shuttle} = 3.0 \text{ ms}^{-1}$ in the direction opposite to that of the exhaust gases.
- 9 **a** $F_{\text{gas}} = ma = m \times \frac{(v - u)}{t}$
 $= 50 \times \left(\frac{180 - 0}{2} \right)$
 $= 4.5 \times 10^3 \text{ N downwards}$
- b** The force of the exhaust gas on the rocket is equal and opposite to the force of the rocket on the exhaust gas.
 $F_{\text{rocket}} = 4.5 \times 10^3 \text{ N upwards}$
- c** Net upwards acceleration = $\frac{\text{resultant force}}{\text{mass}}$
 $= \frac{4.5 \times 10^3 - (225 \times 9.80)}{225}$
 $= 10.2 \text{ ms}^{-2}$
- d** $v = u + at$
 $= (0) + (10.2) \times (2)$
 $= 20.4 \text{ ms}^{-1}$ upwards

Section 8.2 Change in momentum and impulse

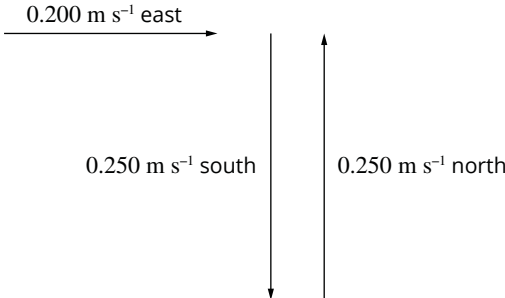
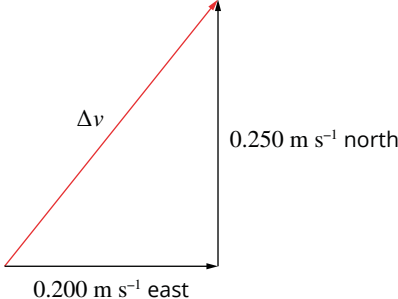
Worked example: Try yourself 8.2.1

IMPULSE OR CHANGE IN MOMENTUM

A student hurries to class after lunch, moving at 4.55 ms^{-1} north. Suddenly the student remembers that she has forgotten her laptop and goes back to her locker at 6.15 ms^{-1} south. If her mass is 75.0 kg , calculate the impulse of the student during the time it takes to turn around.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 75.0 \text{ kg}$ $u = 4.55 \text{ ms}^{-1}$ north $v = 6.15 \text{ ms}^{-1}$ south
Apply the sign convention to the velocity vectors.	$m = 75.0 \text{ kg}$ $u = 4.55 \text{ ms}^{-1}$ $v = -6.15 \text{ ms}^{-1}$
Apply the equation for impulse or change in momentum.	$I = mv - mu$ $= (75.0 \times -6.15) - (75.0 \times 4.55)$ $= -461.25 - 341.25$ $= -802.5 \text{ kg ms}^{-1}$ $= -803 \text{ kg ms}^{-1}$
Apply the sign convention to describe the direction of the impulse.	$I = 803 \text{ kg ms}^{-1}$ south

Worked example: Try yourself 8.2.2
IMPULSE OR CHANGE IN MOMENTUM IN TWO DIMENSIONS

A 65.0 g pool ball is moving at 0.250 m s^{-1} south towards a cushion and bounces off at 0.200 m s^{-1} east. Calculate the impulse on the ball during the change in velocity.

Thinking	Working
Identify the formula for calculating a change in velocity, Δv .	$\Delta v = \text{final velocity} - \text{initial velocity}$
Draw the final velocity, v , and the initial velocity, u , separately. Then draw the initial velocity in the opposite direction, which represents the negative of the initial velocity, $-u$.	
Construct a vector diagram, drawing first v and then from its head draw the opposite of u . The change of velocity vector is drawn from the tail of the final velocity to the head of the opposite of the initial velocity.	
As the two vectors to be added are at 90° to each other, apply Pythagoras' theorem to calculate the magnitude of the change in velocity.	$\Delta v^2 = 0.2^2 + 0.25^2$ $= 0.0400 + 0.0625$ $\Delta v = \sqrt{0.1025}$ $= 0.320 \text{ m s}^{-1}$
Calculate the angle from the north vector to the change in velocity vector.	$\tan \theta = \frac{0.200}{0.250}$ $\theta = \tan^{-1} 0.800$ $= 38.7^\circ$
State the magnitude and direction of the change in velocity.	$\Delta v = 0.320 \text{ m s}^{-1} \text{ N } 38.7^\circ \text{ E}$
Identify the variables using subscripts and ensure that the variables are in their standard units.	$m_1 = 0.0650 \text{ kg}$ $\Delta v = 0.320 \text{ m s}^{-1} \text{ N } 38.7^\circ \text{ E}$
Apply the equation for impulse or change in momentum.	$\Delta p = mv - mu$ $= m(v - u)$ $= m\Delta v$ $= 0.065 \times 0.320$ $= 0.0208 \text{ kg m s}^{-1}$
Apply the direction convention to describe the direction of the change in momentum.	$\Delta p = 0.0208 \text{ kg m s}^{-1} \text{ N } 38.7^\circ \text{ E}$

Section 8.2 Review

KEY QUESTIONS SOLUTIONS

$$\begin{aligned}
 1 \quad I &= mv - mu \\
 &= (9.50 \times -6.25) - (9.50 \times 2.50) \\
 &= -59.375 - 23.75 \\
 &= -83.1 \text{ kg ms}^{-1} \\
 &= 83.1 \text{ kg ms}^{-1} \text{ south}
 \end{aligned}$$

$$\begin{aligned}
 2 \quad I &= mv - mu \\
 &= (6050 \times 16.7) - (6050 \times -22.2) \\
 &= 101\,035 + 134\,310 \\
 &= 235\,000 \text{ kg ms}^{-1} \text{ east}
 \end{aligned}$$

$$\begin{aligned}
 3 \quad \Delta p &= mv - mu \\
 &= (8.00 \times 8.00) - (8.00 \times 3.00) \\
 &= 64.0 - 24.0 \\
 &= 40.0 \text{ kg ms}^{-1} \text{ east}
 \end{aligned}$$

$$\begin{aligned}
 4 \quad \Delta p &= mv - mu \\
 &= (0.250 \times -9.80) - (0.250 \times 0) \\
 &= -2.45 \text{ kg ms}^{-1} \\
 &= 2.45 \text{ kg ms}^{-1} \text{ down}
 \end{aligned}$$

$$\begin{aligned}
 5 \quad \Delta p &= mv - mu \\
 mv &= \Delta p + mu \\
 v &= \frac{\Delta p + mu}{m} \\
 &= \frac{-0.075 + 0.125 \times 3.00}{0.125} \\
 &= 2.4 \text{ ms}^{-1} \text{ north}
 \end{aligned}$$

$$\begin{aligned}
 6 \quad \Delta v &= \text{final velocity} - \text{initial velocity} \\
 &= v - u \\
 &= v + (-u) \\
 &= 45.0 \text{ ms}^{-1} \text{ north} + 45.0 \text{ ms}^{-1} \text{ east}
 \end{aligned}$$

The magnitude of the change in velocity is calculated using Pythagoras' theorem:

$$\begin{aligned}
 \Delta v^2 &= 45.0^2 + 45.0^2 \\
 &= 2025 + 2025
 \end{aligned}$$

$$\begin{aligned}
 \Delta v &= \sqrt{4050} \\
 &= 63.6 \text{ ms}^{-1}
 \end{aligned}$$

Use trigonometry to calculate the angle of the change in momentum.

$$\begin{aligned}
 \tan \theta &= \frac{45.0}{45.0} \\
 \theta &= \tan^{-1}(1) \\
 &= 45^\circ \\
 \Delta v &= 63.6 \text{ ms}^{-1} \text{ N } 45^\circ \text{ E}
 \end{aligned}$$

The magnitude of the change in momentum is calculated using:

$$\begin{aligned}
 \Delta p &= mv - mu \\
 &= m(v - u) \\
 &= m\Delta v \\
 &= 45.0 \times 63.6 \\
 &= 2862 \\
 &= 2860 \text{ kg ms}^{-1} \\
 \Delta p &= 2860 \text{ kg ms}^{-1} \text{ N } 45^\circ \text{ E}
 \end{aligned}$$

7 $\Delta v = \text{final velocity} - \text{initial velocity}$

$$= v - u$$

$$= v + (-u)$$

$$= 3.60 \text{ m s}^{-1} \text{ west} + 4.00 \text{ m s}^{-1} \text{ south}$$

The magnitude of the change in velocity is calculated using Pythagoras' theorem:

$$\Delta v^2 = 3.60^2 + 4.00^2$$

$$= 12.96 + 16.0$$

$$\Delta v = \sqrt{28.9}$$

$$= 5.38 \text{ m s}^{-1}$$

Use trigonometry to calculate the angle of the change in momentum.

$$\tan \theta = \frac{3.60}{4.00}$$

$$\theta = \tan^{-1} (0.9)$$

$$= 42^\circ$$

$$\Delta v = 5.38 \text{ m s}^{-1} \text{ S } 42^\circ \text{ W}$$

The magnitude of the change in momentum is calculated using:

$$\Delta p = mv - mu$$

$$= m(v - u)$$

$$= m\Delta v$$

$$= 70.0 \times 5.38$$

$$= 377 \text{ kg m s}^{-1}$$

$$\Delta p = 377 \text{ kg m s}^{-1} \text{ S } 42^\circ \text{ W}$$

Section 8.3 Newton's first law

Section 8.3 Review

KEY QUESTIONS SOLUTIONS

- 1 The box has changed its velocity so the student can use Newton's first law to conclude that an unbalanced force must have acted on the box to slow it down.
- 2 Even though the car has maintained its speed, the direction has changed, which means the velocity has changed. Using Newton's first law, it can be concluded that an unbalanced force has acted on the car to change its direction.
- 3 B. Since the ball maintains a constant velocity, according to Newton's first law there must not be an unbalanced force. There is no forwards force, friction or air resistance acting on the ball.
- 4 No horizontal force acts on the person. In accordance with Newton's first law of motion, the bus slows, but the standing passenger will continue to move with constant velocity unless acted on by an unbalanced force; usually the passenger will lose his or her footing and fall forwards.
- 5 Constant velocity, so $F_{\text{net}} = 0$, then frictional force = applied force = 20 N.
- 6 **a** Constant velocity, so $F_{\text{net}} = 0$, then frictional force = applied force = 25 N.
b 25 N
c $F \cos 30^\circ = 25 \text{ N}$
 $F = 29 \text{ N}$ at an angle of 30° to the horizontal.
- 7 The plane slows down as it travels along the runway because of the large retarding forces acting on it. The passengers wearing seatbelts would have retarding forces provided by the seatbelt and would slow down at the same rate as the plane. A passenger standing in the aisle, if they were not hanging on to anything, would have no retarding forces acting and so would tend to maintain their original velocity and move towards the front of the plane.
- 8 **a** Gravitational force of attraction between the two masses.
b Electrical force of attraction between the negative electron and the positive nucleus.
c Friction between the tyres and the road.
d Tension in the wire.

- 9 a If the cloth is pulled quickly, the force on the glass acts for a short time only. This force does not overcome the tendency of the glass to stay where it is, i.e. its inertia.
 b Using a full glass makes the trick easier because the force will have less effect on the glass due to its greater mass. The inertia of the full glass is greater than that of an empty glass.
- 10 The fully laden semitrailer will find it most difficult to stop. Its large mass means that more force is required to bring it to a stop.
- 11 Constant speed, so $F_{\text{net}} = 0$ in both vertical and horizontal directions. To exactly balance the other forces, lift = 50 kN up, and drag = 12 kN west.

Section 8.4 Newton's second law

Worked example: Try yourself 8.4.1

CALCULATING THE FORCE THAT CAUSES AN ACCELERATION

Calculate the net force causing a 75.8 kg runner to accelerate at 4.05 m s^{-2} south.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 75.8 \text{ kg}$ $a = 4.05 \text{ m s}^{-2}$ south
Apply the equation for force from Newton's second law.	$F_{\text{net}} = ma$ $= 75.8 \times 4.05$ $= 307 \text{ N}$
Give the direction of the net force, which is the same as the direction of the acceleration.	$F_{\text{net}} = 307 \text{ N south}$

Worked example: Try yourself 8.4.2

CALCULATING THE FINAL VELOCITY OF AN ACCELERATING MASS

Calculate the final velocity of a 307 g fish that accelerates for 5.20 s from rest due to a force of 0.250 N left.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 0.307 \text{ kg}$ $t = 5.20 \text{ s}$ $u = 0 \text{ m s}^{-1}$ $F_{\text{net}} = 0.250 \text{ N left}$
Apply a variations of the equation for force from Newton's second law.	$F_{\text{net}} = m \frac{(v - u)}{t}$ $(v - u) = \frac{(F_{\text{net}} t)}{m}$ $v = \frac{(F_{\text{net}} t)}{m} + u$ $= \frac{0.250 \times 5.20}{0.307} + 0$ $= 4.23 \text{ m s}^{-1}$
Give the direction of the final velocity as being the same as the direction of the force.	$v = 4.23 \text{ m s}^{-1} \text{ left}$

Worked example: Try yourself 8.4.3
CALCULATING THE ACCELERATION OF AN OBJECT WITH MORE THAN ONE FORCE ACTING ON IT

A car with a mass of 900 kg applies a driving force of 3000 N as it starts moving. Friction and air resistance oppose the motion of the car with a force of 750 N. What is the car's initial acceleration?	
Thinking	Working
Determine the individual forces acting on the car, and apply the vector sign convention.	$F_1 = 3000 \text{ N forwards}$ $= 3000 \text{ N}$ $F_2 = 750 \text{ N backwards}$ $= -750 \text{ N}$
Determine the net force acting on the car.	$F_{\text{net}} = F_1 + F_2$ $= 3000 + (-750)$ $= +2250 \text{ N or } 2250 \text{ N forwards}$
Use Newton's second law to determine the acceleration.	$a = \frac{F_{\text{net}}}{m}$ $= \frac{2250}{900}$ $= 2.50 \text{ m s}^{-2} \text{ forwards}$

Worked example: Try yourself 8.4.4
CALCULATING THE ACCELERATION OF A CONNECTED BODY

A 0.6 kg trolley cart is connected by a cord to a 1.5 kg mass. The cord is placed over a pulley and the mass is allowed to fall under the influence of gravity.

a Assuming that the cart can move over the table unhindered by friction, determine the acceleration of the cart.	
Thinking	Working
Recognise that the cart and the falling mass are connected, and determine a sign convention for the motion.	As the mass falls, the cart will move forwards. Therefore, both downwards movement of the mass and forwards movement of the cart will be considered positive motion.
Write down the data that is given. Apply the sign convention to vectors.	$m_1 = 1.5 \text{ kg}$ $m_2 = 0.6 \text{ kg}$ $g = 9.8 \text{ m s}^{-2} \text{ down}$ $= +9.8 \text{ m s}^{-2}$
Determine the forces acting on the system.	The only force acting on the combined system of the cart and mass is the weight of the falling mass. $F_{\text{net}} = F_g$ $= m_1 g$ $= 1.5 \times 9.8$ $= 14.7 \text{ N in the positive direction}$
Calculate the mass being accelerated.	This net force has to accelerate not only the cart but also the falling mass. $m = m_1 + m_2$ $= 1.5 + 0.6$ $= 2.1 \text{ kg}$
Use Newton's second law to determine acceleration.	$a = \frac{F_{\text{net}}}{m}$ $= \frac{14.7}{2.1}$ $= 7.0 \text{ m s}^{-2} \text{ forwards}$

b If a frictional force of 4.2 N acts against the cart, what is the acceleration now?

Thinking

Write down the data that is given. Apply the sign convention to vectors.

Working

$$\begin{aligned} m_1 &= 1.5 \text{ kg} \\ m_2 &= 0.6 \text{ kg} \\ g &= 9.8 \text{ ms}^{-2} \text{ down} \\ &= +9.8 \text{ ms}^{-2} \\ F_{\text{fr}} &= 4.2 \text{ N backwards} \\ &= -4.2 \text{ N} \end{aligned}$$

Determine the forces acting on the system.

There are now two forces acting on the combined system of the cart and mass: the weight of the falling mass and friction.

$$\begin{aligned} F_{\text{net}} &= F_g + F_{\text{fr}} \\ &= 14.7 + (-4.2) \\ &= 10.5 \text{ N} \\ &= 10.5 \text{ N in the positive direction} \end{aligned}$$

Use Newton's second law to determine acceleration.

$$\begin{aligned} a &= \frac{F_{\text{net}}}{m} \\ &= \frac{10.5}{2.1} \\ &= 5.0 \text{ ms}^{-2} \text{ forwards} \end{aligned}$$

Section 8.4 Review

KEY QUESTIONS SOLUTIONS

$$\begin{aligned} 1 \quad a &= \frac{F_{\text{net}}}{m} \\ &= \frac{158}{23.9} \\ &= 6.61 \text{ ms}^{-2} \text{ north} \end{aligned}$$

$$\begin{aligned} 2 \quad m &= \frac{F_{\text{net}}}{a} \\ &= \frac{352}{9.20} \\ &= 38.3 \text{ kg} \end{aligned}$$

$$\begin{aligned} 3 \quad F_{\text{net}} &= \frac{(v - u)t}{t} \\ (v - u) &= \frac{F_{\text{net}}t}{m} \\ v &= \frac{F_{\text{net}}t}{m} + u \\ &= \frac{56.8 \times 3.50}{55.9} + 0 = \frac{56.8 \times 3.50}{55.9} \\ &= 3.56 \text{ ms}^{-2} \text{ north} \end{aligned}$$

$$\begin{aligned} 4 \quad a &= \frac{F_{\text{net}}}{m} \\ &= \frac{441}{45.0} \\ &= 9.80 \text{ ms}^{-2} \text{ down} \end{aligned}$$

$$\begin{aligned} 5 \quad a &= \frac{F_{\text{net}}}{m} \\ &= \frac{882}{90.0} \\ &= 9.8 \text{ ms}^{-2} \text{ down} \end{aligned}$$

$$\begin{aligned}
 6 \quad F_{\text{net}} &= m \frac{(v - u)}{t} \\
 (v - u) &= \frac{F_{\text{net}} t}{m} \\
 v &= \frac{F_{\text{net}} t}{m} + u \\
 &= \frac{-45.5 \times 2.80}{60.0} + 2.67 \\
 &= 0.547 \text{ m s}^{-1} \text{ east}
 \end{aligned}$$

$$\begin{aligned}
 7 \quad a &= \frac{F_{\text{net}}}{m} \\
 &= \frac{95.0}{0.0609} \\
 &= 1560 \text{ m s}^{-2} \text{ south}
 \end{aligned}$$

$$\begin{aligned}
 8 \quad m &= \frac{F_{\text{net}}}{a} \\
 &= \frac{565000}{7.20} \\
 &= 78500 \text{ kg}
 \end{aligned}$$

$$\begin{aligned}
 9 \quad F_{\text{net}} &= m \frac{(v - u)}{t} \\
 (v - u) &= \frac{F_{\text{net}} t}{m} \\
 v &= \frac{F_{\text{net}} t}{m} + u \\
 &= \frac{0.0823 \times 0.0105}{0.003} + 0 \\
 &= 0.288 \text{ m s}^{-1} \text{ north}
 \end{aligned}$$

$$\begin{aligned}
 10 \text{ a } \Sigma F_{\text{h}} &= 45 \text{ N south} + 25 \text{ N north} \\
 &= -45 + 25 \\
 &= -20 \\
 &= 20 \text{ N south}
 \end{aligned}$$

$$\begin{aligned}
 \text{b } F_{\text{net}} &= ma \\
 20 &= 65a \\
 a &= 0.31 \text{ m s}^{-2}
 \end{aligned}$$

$$\begin{aligned}
 11 \text{ a } F_{\text{net}} &= F_{\text{g}} = mg = 0.50 \times 9.8 = 4.9 \text{ N} \\
 a &= \frac{F_{\text{net}}}{m} = \frac{4.9}{(2.5 + 0.5)} = 1.6 \text{ m s}^{-2}
 \end{aligned}$$

$$\text{b } v = u + at = 0 + 1.6 \times 0.5 = 0.8 \text{ m s}^{-1}$$

$$\begin{aligned}
 \text{c } F_{\text{net}} &= ma = F_{\text{g}} - F_{\text{f}} \\
 \text{so } F_{\text{net}} &= 4.9 - 4.3 = 0.6 \text{ N} \\
 \text{and } a &= \frac{F_{\text{net}}}{m} = \frac{0.6}{3} = 0.2 \text{ m s}^{-2}
 \end{aligned}$$

12 Determine the driving force provided by the truck based on the information for when it is empty.

$$\begin{aligned}
 F_{\text{net}} &= ma \\
 &= 2000 \times 2.0 \\
 &= 4000 \text{ N}
 \end{aligned}$$

Calculate the total mass of the truck for the new acceleration.

$$\begin{aligned}
 m &= \frac{F_{\text{net}}}{a} \\
 &= \frac{4000}{1.25} \\
 &= 3200 \text{ kg}
 \end{aligned}$$

So the mass of the boxes must be:

$$3200 - 2000 = 1200 \text{ kg}$$

$$\begin{aligned}
 \text{Number of boxes} &= \frac{1200}{300} \\
 &= 4 \text{ boxes}
 \end{aligned}$$

$$\begin{aligned}
 13 \quad F_{\text{net}} &= \text{thrust} - \text{weight of rocket} \\
 &= 1\,000\,000 - (50\,000 \times 9.8) \\
 &= 510\,000 \text{ N} \\
 a &= \frac{F_{\text{net}}}{m} \\
 &= \frac{510\,000}{50\,000} \\
 &= 10.2 \text{ ms}^{-2}
 \end{aligned}$$

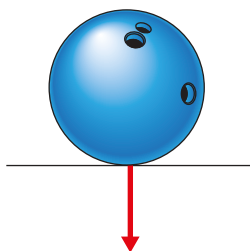
Section 8.5 Newton's third law

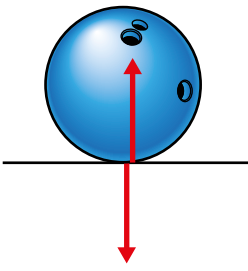
Worked example: Try yourself 8.5.1

APPLYING NEWTON'S THIRD LAW

In the diagram below, a bowling ball is resting on the floor and one of the forces is given. Copy the diagram into your book and complete the following:

- Label the given force using the system ' F on _____ by _____'.
- Label the reaction force to the given force using the system ' F on _____ by _____'.
- Draw the reaction force on the diagram, showing its size and location.



Thinking	Working
Identify the two objects involved in the action–reaction pair.	The bowling ball and the floor.
Identify which object is applying the force and which object is experiencing the force, for the force vector shown.	The force vector shown is a force from the bowling ball on the floor.
Use the system of labelling action and reaction forces ' F_{on} _____ by _____', to label the action force.	$F_{\text{on floor by bowling ball}}$
Use the system of labelling action and reaction forces ' F_{on} _____ by _____', to label the reaction force.	$F_{\text{on bowling ball by floor}}$
Use a ruler to measure the length of the action force and construct a vector arrow in the opposite direction with its tail on the point of application of the reaction force.	

Section 8.5 Review

KEY QUESTIONS SOLUTIONS

- There is a force on the hammer by the nail, and a force on the nail by the hammer. These two forces are equal in magnitude and opposite in direction.
- The force arrow shown is the $F_{\text{on the astronaut by the Earth}}$
 - The reaction force must act on the other object, so in this case it is the $F_{\text{on the Earth by the astronaut}}$
- The force on the hand by the water. The swimmer creates the action force by pushing on the water, but the reaction force acts on the swimmer which moves him in the direction of his motion.
- The force on the balloon by the escaping air. The balloon's elasticity compresses the air inside and pushes it out of the mouth of the balloon. This is the action force. The air must therefore exert an equal and opposite forwards force on the balloon, which in turn moves the balloon around the room.
- The reaction force is on the racquet by the ball, resulting in a force of 100 N east.
- The boat exerts an equal and opposite reaction force, i.e. 140 N in the opposite direction to the leaping fisherman.
 - $a = \frac{F_{\text{net}}}{m} = \frac{140}{40} = 3.5 \text{ ms}^{-2}$ in the opposite direction to the fisherman
 - Acceleration of the fisherman: $a = \frac{F_{\text{net}}}{m} = \frac{140}{70} = 2.0 \text{ ms}^{-2}$
Speed of the fisherman: $v = 0 + 2 \times 0.5 = 1.0 \text{ ms}^{-1}$
Speed of the boat: $v = 0 + 3.5 \times 0.5 = 1.8 \text{ ms}^{-1}$
- The astronaut should throw the tool kit in the opposite direction from the ship. By throwing the tool kit, there is an action force on the tool kit by the astronaut which is directed away from the ship. According to Newton's third law, there will be a reaction force on the astronaut by the tool kit that will be in the opposite direction, i.e. towards the ship.
- Tania is correct. For an action–reaction pair, the action force is a force on object A by object B, and the reaction force is a force on object B by object A. That is, the two forces act on different objects. In this case, both the weight force and the normal force are acting on the same object: the lunch box.

Section 8.6 Impulse and force

Worked example: Try yourself 8.6.1

CALCULATING THE FORCE AND IMPULSE

A student drops a 56.0 g egg onto a table from a height of 60 cm. Just before the egg hits the table, the velocity of the egg is 3.43 ms^{-1} down. The egg's final velocity is zero as it smashes on the table. The time it takes for the egg to change its velocity to zero is 3.55 ms.

a Calculate the change in momentum of the egg.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 0.0560 \text{ kg}$ $u = 3.43 \text{ ms}^{-1}$ down $v = 0 \text{ ms}^{-1}$
Apply the sign and direction convention for motion in one dimension. Up is positive and down is negative.	$m = 0.0560 \text{ kg}$ $u = -3.43 \text{ ms}^{-1}$ $v = 0 \text{ ms}^{-1}$
Apply the equation for change in momentum.	$\Delta p = m(v - u)$ $= 0.0560 \times (0 - (-3.43))$ $= 0.192 \text{ kg ms}^{-1}$
Refer to the sign and direction convention to determine the direction of the change in momentum.	$\Delta p = 0.192 \text{ kg ms}^{-1}$ up

b Calculate the impulse of the egg.	
Thinking	Working
Using the answer to part (a), apply the equation for impulse.	$I = \Delta p$ $= 0.192 \text{ kg ms}^{-1}$
Refer to the sign and direction convention to determine the direction of the impulse.	$I = 0.192 \text{ kg ms}^{-1}$ up
c Calculate the average force that acts to cause the impulse.	
Thinking	Working
Use the answer to part (b). Ensure that the variables are in their standard units.	$I = 0.192 \text{ kg ms}^{-1}$ $\Delta t = 3.55 \times 10^{-3} \text{ s}$
Apply the equation for force.	$F\Delta t = I$ $F = \frac{I}{\Delta t}$ $= \frac{0.192}{3.55 \times 10^{-3}}$ $= 54.1 \text{ N}$
Refer to the sign and direction convention to determine the direction of the force.	$F = 54.1 \text{ N}$ up

Worked example: Try yourself 8.6.2

CALCULATING THE FORCE AND IMPULSE (SOFT LANDING)

A student drops a 56.0 g egg into a mound of flour from a height of 60 cm. Just before the egg hits the mound of flour, the velocity of the egg is 3.43 ms^{-1} down. The egg's final velocity is zero as it sinks into the mound of flour. The time it takes for the egg to change its velocity to zero is 0.325 s.

a Calculate the change in momentum of the egg.	
Thinking	Working
Ensure that the variables are in their standard units.	$m = 0.0560 \text{ kg}$ $u = 3.43 \text{ ms}^{-1}$ down $v = 0 \text{ ms}^{-1}$
Apply the sign and direction convention for motion in one dimension. Up is positive and down is negative.	$m = 0.0560 \text{ kg}$ $u = -3.43 \text{ ms}^{-1}$ $v = 0 \text{ ms}^{-1}$
Apply the equation for change in momentum.	$\Delta p = m(v - u)$ $= 0.0560 \times (0 - (-3.43))$ $= 0.192 \text{ kg ms}^{-1}$
Refer to the sign and direction convention to determine the direction of the change in momentum.	$\Delta p = 0.192 \text{ kg ms}^{-1}$ up
b Calculate the impulse of the egg.	
Thinking	Working
Using the answer to part (a), apply the equation for impulse.	$I = \Delta p$ $= 0.192 \text{ kg ms}^{-1}$
Refer to the sign and direction convention to determine the direction of the impulse.	$I = 0.192 \text{ kg ms}^{-1}$ up

c Calculate the average force that acts to cause the impulse.

Thinking

Using the answer to part (b), ensure that the variables are in their standard units.

Apply the equation for force.

Refer to the sign and direction convention to determine the direction of the force.

Working

$$I = 0.192 \text{ kg ms}^{-1}$$

$$\Delta t = 0.325 \text{ s}$$

$$F \Delta t = I$$

$$F = \frac{I}{\Delta t}$$

$$= \frac{0.192}{0.325}$$

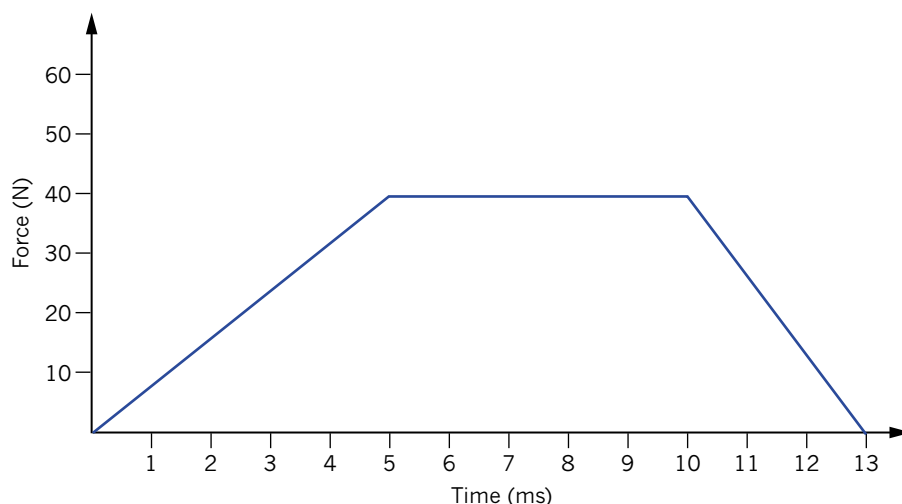
$$= 0.591 \text{ N}$$

$$F = 0.591 \text{ N up}$$

Worked example: Try yourself 8.6.3

CALCULATING THE TOTAL IMPULSE FROM A CHANGING FORCE

A student records the force acting on a tennis ball as it bounces off a hard concrete floor over a period of time. The graph shows the forces acting on a ball during its collision with the concrete floor.

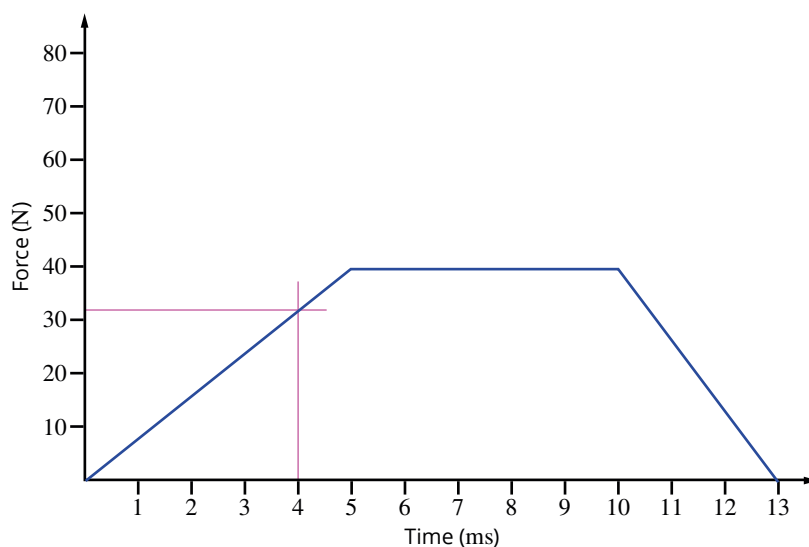


a Determine the force acting on the ball at a time of 4.0 ms.

Thinking

From the 4.0 ms point on the x-axis go up to the line of the graph, then across to the y-axis.

Working



The force is estimated by reading the intercept of the y-axis

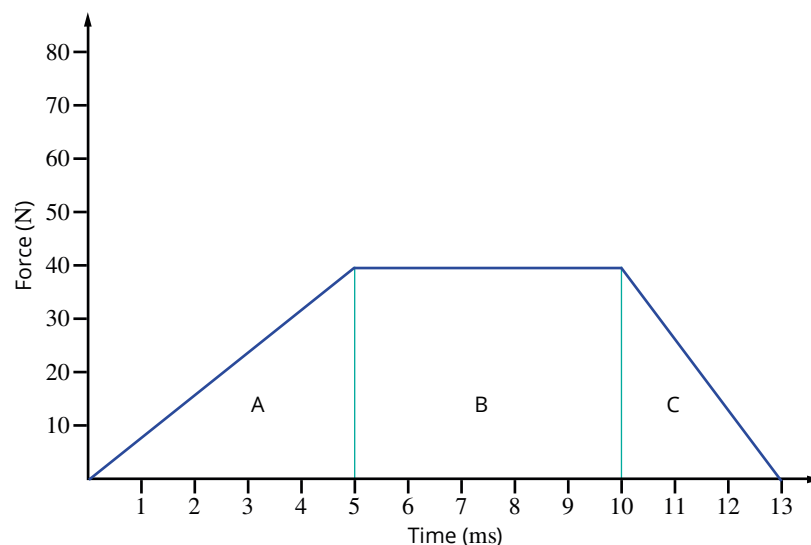
$$F = 32 \text{ N}$$

b Calculate the total impulse of the ball over the 13 ms period of time.

Thinking

Break the area under the graph into sections for which you can calculate the area.

In this case, the graph can be broken into three sections: A, B and C.



Calculate the area of the three sections A, B and C using the equations for the area of a triangle and the area of a rectangle.

$$\begin{aligned}
 \text{area} &= A + B + C \\
 &= \left(\frac{1}{2} b \times h \right) + (b \times h) + \left(\frac{1}{2} b \times h \right) \\
 &= \left[\frac{1}{2} \times (5 \times 10^{-3}) \times 40 \right] + [(5 \times 10^{-3}) \times 40] + \left[\frac{1}{2} \times (3 \times 10^{-3}) \times 40 \right] \\
 &= 0.1 + 0.2 + 0.06 \\
 &= 0.36
 \end{aligned}$$

The total impulse is equal to the area.

$$\begin{aligned}
 I &= \text{area} \\
 &= 0.36 \text{ kg m s}^{-1}
 \end{aligned}$$

Apply the sign and direction convention for motion in one dimension vertically.

$$I = 0.36 \text{ kg m s}^{-1} \text{ up}$$

Section 8.6 Review

KEY QUESTIONS SOLUTIONS

$$\begin{aligned}
 1 \quad a \quad \Delta p &= m(v - u) \\
 &= 45.0 \times (12.5 - 2.45) \\
 &= 452 \text{ kg m s}^{-1} \text{ east}
 \end{aligned}$$

$$\begin{aligned}
 b \quad I &= \Delta p \\
 &= 452 \text{ kg m s}^{-1} \text{ east}
 \end{aligned}$$

$$\begin{aligned}
 c \quad F_{av} \Delta t &= I \\
 F_{av} &= \frac{I}{\Delta t} \\
 &= \frac{452}{3.50} \\
 &= 129 \text{ N east}
 \end{aligned}$$

- 2 Airbags are designed to increase the duration of the collision, which changes the momentum of a person's head during a car accident. Increasing the duration of the collision decreases the force, which reduces the severity of injury.

- 3 a** $\Delta p = m(v - u)$
 $= 0.075 \times (0 - (-15.6))$
 $= 1.17 \text{ kg m s}^{-1} \text{ east}$
- b** $I = \Delta p$
 $= 1.17 \text{ kg m s}^{-1} \text{ east}$
- c** $F_{\text{av}} \Delta t = I$
 $F_{\text{av}} = \frac{I}{\Delta t}$
 $= \frac{1.17}{0.100}$
 $= 11.7 \text{ N east}$
- 4** $F_{\text{av}} \Delta t = I$
 $F_{\text{av}} = \frac{I}{\Delta t}$
 $= \frac{1.17}{0.300}$
 $= 3.90 \text{ N east}$
- 5 a** Impulse $= F \Delta t = \Delta p$
 $= 0.200 \times 45$
 $= 9.0 \text{ kg m s}^{-1}$
- b** $F = \frac{9.0 \text{ kg m s}^{-1}}{0.05 \text{ s}}$
 $= 180 \text{ N in the direction of the ball's travel}$
- c** 180 N in the opposite direction to the ball's travel.
- 6 a** Maximum force $= 1200 \text{ N}$
- b** Impulse $= F \Delta t = \text{area under force-time graph} = 63 \text{ N s}$
- 7 a** $\Delta p = m(v - u)$
 $= (0.025) \times (0 - 50)$
 $= 1.25 \text{ kg m s}^{-1} \text{ opposite in direction to its initial velocity}$
- b** Impulse $= F \Delta t = \Delta p$
 $= 1.25 \text{ kg m s}^{-1} \text{ opposite in direction to its initial velocity}$
- c** $v^2 = u^2 + 2as$
 $0 = 50^2 + 2a(2.0 \times 10^{-2})$
 $a = -6.25 \times 10^4 \text{ m s}^{-2}$
 $F = ma$
 $= 0.025(-6.25 \times 10^4)$
 $= 1.6 \times 10^3 \text{ N in the opposite direction to the initial velocity of the arrow}$
- 8 a** The crash helmet is designed so that the stopping time is increased by the collapsing shell during impact. This will reduce the force, as impulse $= F \Delta t = \Delta p$.
- b** No. A rigid shell would reduce the stopping time, therefore increasing the force.

Section 8.7 Mass and weight

Section 8.7 Review

KEY QUESTIONS SOLUTIONS

- 1 50 kg. The mass of an object does not depend on its environment.
- 2 60 kg is the student's mass, not her weight. Weight is a force and so it is measured in newtons. On Earth, her weight will be 9.8 times larger than her mass.
- 3 $F_g = mg = 75 \times 9.8 = 735 \text{ N}$
- 4 $m = \frac{F_g}{g}$
 $= \frac{34.3}{9.8}$
 $= 3.5 \text{ kg}$
- 5 $F_g = mg$
 $= 3.5 \times 1.6$
 $= 5.6 \text{ N}$
- 6 The mass of the hammer remains constant at 1.5 kg.
 The weight of the hammer on Mars is $F_g = mg = 1.5 \times 3.6 = 5.4 \text{ N}$.
- 7 The weight of any object will be less on the Moon compared with its weight on the Earth as gravity is weaker on the Moon, due to its smaller mass.

CHAPTER 8 REVIEW

- 1 No, a force has not pushed the passengers backwards. Since the passengers have inertia, as the train has started moving forwards the passengers' masses resist the change in motion. According to Newton's first law, their bodies are simply maintaining their original state of being motionless until an unbalanced force acts to accelerate them.
- 2 D. An object travelling at a constant velocity will do so without any force acting.
- 3 $m = \frac{F_{\text{net}}}{a}$
 $= \frac{352}{9.20}$
 $= 38.3 \text{ kg}$
- 4 $a = \frac{F_{\text{net}}}{m}$
 $= \frac{3550}{657}$
 $= 5.40 \text{ ms}^{-2} \text{ north}$
- 5 $a = \frac{F_{\text{net}}}{m}$
 $= \frac{150}{100}$
 $= 1.5 \text{ ms}^{-2}$
- 6 $F_{\text{net}} = 150 - 45$
 $= 105 \text{ N}$
 $a = \frac{F_{\text{net}}}{m}$
 $= \frac{105}{100}$
 $= 1.05 \text{ ms}^{-2}$

7 Using Newton's second law:

$$\begin{aligned} F_{\text{net}} &= ma \\ &= 100 \times 0.6 \\ &= 60.0 \text{ N} \end{aligned}$$

Net force is also given by the sum of individual forces:

$$\begin{aligned} F_{\text{net}} &= 150 - F_{\text{fr}} \\ F_{\text{fr}} &= 150 - F_{\text{net}} \\ &= 150 - 60 \\ &= 90 \text{ N} \end{aligned}$$

8 Using Newton's second law:

$$\begin{aligned} F_{\text{net}} &= ma \\ &= 125 \times 0.800 \\ &= 100 \text{ N} \end{aligned}$$

Net force is also given by the sum of individual forces:

$$\begin{aligned} F_{\text{net}} &= F_{\text{forwards}} - 30 \\ F_{\text{forwards}} &= F_{\text{net}} + 30 \\ &= 100 + 30 \\ &= 130 \text{ N} \end{aligned}$$

9 $F = m \frac{(v - u)}{\Delta t}$

$$\begin{aligned} (v - u) &= \frac{F \Delta t}{m} \\ v &= \frac{F \Delta t}{m} + u \\ &= \frac{-62.0 \times 2.00}{4.0} + 3.75 \\ &= -5.11 \text{ ms}^{-1} \end{aligned}$$

$\therefore v = 5.11 \text{ ms}^{-1}$ west

10 Newton's third law states that every action has an equal and opposite reaction. Therefore, the reaction of the board acting on the student results in a force of 75.0 N north.

$$\begin{aligned} \Delta p &= m(v - u) \\ &= 155 \times (3.25 - 6.50) \\ &= 504 \text{ kg ms}^{-1} \text{ west} \end{aligned}$$

$$\begin{aligned} \Delta p &= mv - mu \\ &= (25.5 \times -2.25) - (25.5 \times 6.40) \\ &= -221 \text{ kg ms}^{-1} \\ &= 221 \text{ kg ms}^{-1} \text{ backwards} \end{aligned}$$

13 $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$

$$\begin{aligned} m_1 u_1 + m_2 u_2 &= m_1 v_1 + m_2 v_2 \\ (40.0 \times 0) + (154 \times 0) &= (40.0 \times 2.15) + 154 v_2 \\ 154 v_2 &= 0 + -86 \\ v_2 &= \frac{-86}{154} \\ &= -0.558 \text{ ms}^{-1} \end{aligned}$$

The astronaut moves backwards at 0.558 ms^{-1} .

- 14** $\Delta v = \text{final velocity} - \text{initial velocity}$

$$= v - u$$

$$= v + (-u)$$

$$= 5.00 \text{ ms}^{-1} \text{ north} + 4.00 \text{ ms}^{-1} \text{ east}$$

The magnitude of the change in velocity is calculated using Pythagoras' theorem:

$$\Delta v^2 = 5.00^2 + 4.00^2$$

$$= 25.0 + 16.0$$

$$\Delta v = \sqrt{41.0}$$

$$= 6.40 \text{ ms}^{-1}$$

Use trigonometry to calculate the angle of the change in momentum.

$$\tan \theta = \frac{4.00}{5.00}$$

$$\theta = \tan^{-1}(0.9)$$

$$= 38.7^\circ$$

$$\Delta v = 6.40 \text{ ms}^{-1} \text{ N } 38.7^\circ \text{ E}$$

The magnitude of the change in momentum is calculated using:

$$\Delta p = mv - mu$$

$$= m(v - u)$$

$$= m\Delta v$$

$$= 75.0 \times 6.40$$

$$= 480 \text{ kg ms}^{-1}$$

$$= 480 \text{ kg ms}^{-1} \text{ N } 38.7^\circ \text{ E}$$

- 15** $\Delta p = m(v - u)$

$$= 0.300 \times (0 - (-5.60))$$

$$= 1.68 \text{ kg ms}^{-1} \text{ east}$$

$$F_{av} \Delta t = \Delta p$$

$$F_{av} = \frac{\Delta p}{\Delta t}$$

$$= \frac{1.68}{1.00}$$

$$= 1.68 \text{ N east}$$

- 16** When a car travelling at a very fast speed comes to a complete halt, the occupants will experience a force applied to them of $F_{\text{net}} = ma = \frac{I}{\Delta t}$. In this case, the mass of the occupants and the impulse (Δp) is always the same. The amount

of force acting upon the occupants therefore depends entirely upon the value of their acceleration, or the time over which a collision takes place; a short collision will involve a large deceleration and therefore a large force applied to the occupants. By designing the bonnet of the car to be long and to crumple, a collision will deform this metal and slow the car down before the impact reaches the occupants, thereby reducing the total force applied to the occupants.

A frame made out of metal that is of medium rigidity is best for this purpose. If the metal of the car frame crumpled too easily then it would not slow down the collision very much, and the force applied to occupants would be high. If the car frame remained too rigid in the event of a collision, then none of the energy of the crash would go into deforming the metal, and the force would be passed entirely to the car occupants. The ideal metal shell should be as strong as possible, so that it would still crumple in the event of a collision. Car manufacturers subject their cars to crash testing during their design in order to optimise this collision time.

- 17 a** $p = mv$

$$pi = 70.0 \times 5.0$$

$$= 350 \text{ kg ms}^{-1}$$

- b** $I = \Delta p$

$$= (0 - 350)$$

$$= -350 \text{ kg ms}^{-1}$$

$$F_{\text{net}} = \frac{I}{\Delta t}$$

$$= \frac{-350}{0.350}$$

$$= -1000 \text{ N}$$

Hence 1000 N of force would act on Young.

$$\begin{aligned} \text{c } F_{\text{net}} &= \frac{I}{\Delta t} \\ &= \frac{-350}{7.00 \times 10^{-3}} \\ &= -50\,000 \text{ N} \end{aligned}$$

Hence 50 000 N of force would act on Young's head. Clearly a helmet is important in this case.

- d** As well as reducing the force acting on the user's head, a crash helmet spreads out the area of the head over which the force is applied, which reduces the risk of penetration of the skull.

18 $\Delta p = \text{impulse} = \text{area under } F-t \text{ graph} = 0.5 \times 0.04 \times 500 = 10 \text{ kg ms}^{-1}$

- 19** As the bat and ball form an isolated system, momentum is conserved. The gain in momentum of the ball is equal to the loss of momentum of the bat. Hence:

$$\Delta p = 10 \text{ kg ms}^{-1}$$

20 $\Delta p = m\Delta v$

Therefore:

$$\begin{aligned} \Delta v &= \frac{\Delta p}{m} \\ &= \frac{10}{0.170} \\ &= 58.8 \\ &= 59 \text{ ms}^{-1} \end{aligned}$$

21 $F_g = mg = 10 \times 9.8 = 98 \text{ N}$

22 $m = \frac{F_g}{g}$
 $= \frac{20.6}{9.8}$
 $= 2.1 \text{ kg}$

23 a mass = 85 kg

b mass = 85 kg

c $F_g = mg = 85 \times 3.6 = 306 \text{ N down}$

- 24** The object's greatest weight is when it is on Earth. The second greatest weight is on Mars, and its least weight is when it is on the Moon.